

Two-dimensional Random variables

Let S be the sample space associated with a random experiment. Let $X = X(s)$ and $Y = Y(s)$ be two functions each assigning a real no. to each $s \in S$. Then (X, Y) is called a two dimensional RV.

- * If the possible values of (X, Y) are finite or countable infinite, (X, Y) is called a two-dimensional discrete RV.
- * If (X, Y) can assume all values in a special region R in the XY -plane, then (X, Y) is called a two-dimensional continuous RV.

Joint probability function of (X, Y)

If (X, Y) is a two-dim discrete random variable then joint probability mass function of X, Y is $P(X = x_i, Y = y_j)$, then P_{ij} or P_{x_i, y_j} is called the probability mass function or simply the probability function of (X, Y) provided the following conditions are satisfied.

(i) $P_{ij} \geq 0 \quad \forall i, j$

(ii) $\sum_j \sum_i P_{ij} = 1$

The set of triples (x_i, y_j, p_{ij}) ; $i = 1, 2, \dots, m$
 $j = 1, 2, \dots, n$
is called the joint probability distribution of (X, Y) .

X \ Y	1	2	3	4	5	6	Total
1	$\frac{1}{6}$	0	0	0	0	0	$\frac{1}{6}$
2	$\frac{1}{36}$	$\frac{5}{36}$	0	0	0	0	$\frac{6}{36}$
3	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{4}{36}$	0	0	0	$\frac{6}{36}$
4	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{3}{36}$	0	0	$\frac{6}{36}$
5	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{2}{36}$	0	$\frac{6}{36}$
6	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
Total	$\frac{16}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$	1

$p_{x,y}(1,1) = P(X=1, Y=1)$
 $= \frac{6}{36} = \frac{1}{6}$

$p_{x,y}(1,2) = P(X=1, Y=2) = 0$

$p_{x,y}(1,3) = P(X=1, Y=3) = 0$

$p_{x,y}(2,1) = P(X=2, Y=1) = \frac{1}{36}$

$p_{x,y}(2,2) = P(X=2, Y=2) = \frac{5}{36}$

here $P_{ij} \geq 0$

and $\sum_{j=1}^n \sum_{i=1}^m P_{ij} = 1$

Suppose $p_{x,y}(2,5,3) = 0$
Since $X=2, Y=5$ is not in the sample space

$$p_{x,y}(x_i, y_j) = \begin{cases} P(X=x_i, Y=y_j) & \text{for a value } (x_i, y_j) \in (X, Y) \\ 0 & \text{otherwise} \end{cases}$$

Let $X(s) = \{x_1, x_2, \dots, x_n\}$

and $Y(s) = \{y_1, y_2, \dots, y_m\}$

$X \times Y = \{x_1, x_2, \dots, x_n\} \times \{y_1, y_2, \dots, y_m\}$

Then $p_{x,y}(x_i, y_j) = P(X=x_i, Y=y_j) = P_{ij}$ can be represented by the following table

X \ Y	y_1	y_2	y_m	Total
x_1	p_{11}	p_{12}	p_{1m}	$p_{1.}$
x_2	p_{21}	p_{22}	p_{2m}	$p_{2.}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	p_{n1}	p_{n2}	p_{nm}	$p_{n.}$
Total	$p_{.1}$	$p_{.2}$	$p_{.m}$	1

Ex: A pair of fair dice are thrown.
Let X denote the no. of the first die
and Y the smaller of the two numbers.

Here $S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
 $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$
 $(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

Joint probability density function

If (X, Y) is a two-dimensional continuous RV then

$$P\left\{x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2} \text{ and } y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right\} = f(x, y) dx dy$$

where $f(x, y)$ is a function of x and y so that $f(x, y) dx dy$ represents the probability that the two-dim continuous RV (X, Y) lie in the infinitesimal rectangular region

$$\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}, y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right)$$

of area $dx dy$. The function $f(x, y)$ is called joint probability density function of X and Y and it satisfies the following conditions

(i) $f(x, y) \geq 0, \quad \forall (x, y) \in R$, where R is the range space.

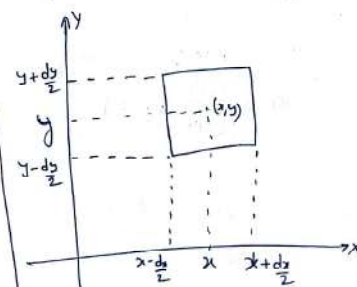
(ii) $\iint_R f(x, y) dx dy = 1$

Moreover, if D is a subspace of the range space R , $P\{(X, Y) \in D\}$ is defined

$$P\{(X, Y) \in D\} = \iint_D f(x, y) dx dy$$

In particular

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dx dy$$



$$P(a \leq x \leq b, c \leq y \leq d) = \int_c^d \int_a^b f_{xy}(x, y) dx dy$$

Cumulative Distribution function (CDF)

If (X, Y) is a two-dimensional RV, then $F(x, y) = P\{X \leq x \text{ and } Y \leq y\}$ is called the cdf of (X, Y)

In discrete case,

$$F(x, y) = \sum_{j: y_j \leq y} \sum_{i: x_i \leq x} p_{ij}$$

In the continuous case

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

Properties of $F(x, y)$

- (i) $F(-\infty, y) = 0 = F(x, -\infty)$ and $F(\infty, \infty) = 1$
- (ii) $P\{a < X \leq b, Y \leq y\} = F(b, y) - F(a, y)$
- (iii) $P\{X \leq x, c < Y \leq d\} = F(x, d) - F(x, c)$
- (iv) $P\{a < X \leq b, c < Y \leq d\} = F(b, d) - F(a, d) - F(b, c) + F(a, c)$
- (v) At pt. of continuity of $f(x, y)$

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y)$$

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Marginal probability Distribution (discrete)

Let (X, Y) be a discrete two-dimensional R.V. which takes up countable no. of values (x_i, y_j) . Then the probability distribution of X is denoted as follows
 $p(x_i) = P(X = x_i) = P(X = x_i \cap Y = y_1) + P(X = x_i \cap Y = y_2) + \dots + P(X = x_i \cap Y = y_m)$

$$= p_{i1} + p_{i2} + \dots + p_{im} = \sum_{j=1}^m p_{ij}$$

and is known as marginal probability function of X .

Similarly,

$$\sum_{i=1}^n p_{ij} = p_{j1} + p_{j2} + \dots + p_{jn} = \sum_{i=1}^n \sum_{j=1}^m p_{ij} = 1$$

Similarly $p(y_j) = P(Y = y_j) = \sum_{i=1}^n p_{ij}$ which is the marginal probability mass function of Y .

Ex Example of pair of dice

$$p(x=1) = P(X=1)$$

$$= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) + P(X=1, Y=4) + P(X=1, Y=5) + P(X=1, Y=6)$$

$$= \frac{1}{6} + 0 = \frac{1}{6}$$

i.e. marginal probability of $X=1$.

$$\text{Similarly } p(X=2) = P(X=2) = \frac{1}{6}$$

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Marginal probability function (CRV) Marginal probability density function

The marginal p.d.f. of RV X & Y are denoted by $f_x(x)$ and $f_y(y)$ respectively and defined as

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, y) dy$$

$$\text{and } f_y(y) = \int_{-\infty}^{\infty} f_{xy}(x, y) dx$$

In case of two-dim continuous RV, the marginal distribution function are defined as

$$F_x(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{xy}(x, y) dy \right\} dx = P(X \leq x)$$

$$\text{and } F_y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{xy}(x, y) dx \right\} dy = P(Y \leq y)$$

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Conditional Probability Distribution

$$P(X = x_i / Y = y_j) = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{ij}}{p_{.j}}$$

is called the conditional probability function of X , given that $Y = y_j$

The collection of pairs, $\{x_i, \frac{p_{ij}}{p_{.j}}\}$, $i=1, 2, \dots$

is called the conditional probability distribution of X , given $Y = y_j$

Similarly, the collection of pairs, $\{y_j, \frac{p_{ij}}{p_{i.}}\}$, $i=1, 2, \dots$ is called the conditional probability of Y , given $X = x_i$.

For continuous case

$$P\left\{x - \frac{dx}{2} \leq X < x + \frac{dx}{2} / Y = y\right\} =$$

$$P\left\{x - \frac{dx}{2} \leq X < x + \frac{dx}{2} / y - \frac{dy}{2} \leq Y < y + \frac{dy}{2}\right\}$$

$$= \frac{f(x, y) dx dy}{f_y(y) dy} = \left\{ \frac{f(x, y)}{f_y(y)} \right\} dx$$

$\frac{f(x, y)}{f_y(y)}$ is called the conditional density of X , given Y , and is denoted by $f(x/y)$

Similarly, $\frac{f(x, y)}{f_x(x)}$ is called the conditional density of Y , given X , and is denoted by $f(y/x)$

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Ex For the joint probability distribution of two RV X & Y given below

$X \backslash Y$	1	2	3	4	Total
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find ① the marginal distribution of X and Y .

② Conditional distribution of X given the value of $Y=1$ and that of Y given the value of $X=2$.

Soln ① Marginal distribution of $X \rightarrow P(X=x) = \sum_y P(X=x, Y=y)$

X	1	2	3	4
$P(X)$	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

marginal distribution of $Y \rightarrow P(Y=y) = \sum_x P(X=x, Y=y)$

Y	1	2	3	4
$P(Y)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

② Conditional probability funcⁿ of X given $Y=1$

$$P(X=x | Y=1) = \frac{P(X=x, Y=1)}{P(Y=1)}$$

$$P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{\frac{4}{36}}{\frac{11}{36}} = \frac{4}{11}$$

$$P(X=2 | Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{\frac{3}{36}}{\frac{11}{36}} = \frac{3}{11}$$

$$P(X=3 | Y=1) = \frac{P(X=3, Y=1)}{P(Y=1)} = \frac{\frac{2}{36}}{\frac{11}{36}} = \frac{2}{11}$$

$$P(X=4 | Y=1) = \frac{P(X=4, Y=1)}{P(Y=1)} = \frac{\frac{1}{36}}{\frac{11}{36}} = \frac{1}{11}$$

$\therefore$ Conditional distribution of X given $Y=1$

$X=2$	1	2	3	4
$P(X=2 Y=1)$	$\frac{4}{11}$	$\frac{3}{11}$	$\frac{2}{11}$	$\frac{1}{11}$

Conditional probability function of Y given $X=2$

$$\frac{P(X=2, Y=y)}{P(X=2)}$$

$$P(Y=1 | X=2) = \frac{P(X=2, Y=1)}{P(X=2)} = \frac{\frac{1}{36}}{\frac{9}{36}} = \frac{1}{9}$$

$$P(Y=2 | X=2) = \frac{P(X=2, Y=2)}{P(X=2)} = \frac{\frac{3}{36}}{\frac{9}{36}} = \frac{3}{9} = \frac{1}{3}$$

$$P(Y=3 | X=2) = \frac{P(X=2, Y=3)}{P(X=2)} = \frac{\frac{3}{36}}{\frac{9}{36}} = \frac{1}{3}$$

$$P(Y=4 | X=2) = \frac{P(X=2, Y=4)}{P(X=2)} = \frac{\frac{2}{36}}{\frac{9}{36}} = \frac{2}{9}$$

$\therefore$ Conditional distribution of Y given $X=2$

$Y=y$	1	2	3	4
$P(Y=y X=2)$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{9}$

Ex For the bivariate probability distribution of X and Y

find

① $P(X \leq 1, Y=2)$

② $P(X \leq 1)$

③ $P(Y \leq 3)$

④ $P(X < 3, Y \leq 4)$

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Soln Marginal distribution

$X \backslash Y$	1	2	3	4	5	6	P_{i*}
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
P_{*j}	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	1

① $P(X \leq 1, Y=2)$

$$= P(X=0, Y=2) + P(X=1, Y=2)$$

$$= 0 + \frac{1}{16} = \frac{1}{16}$$

② $P(X \leq 1) = P(X=0) + P(X=1)$

$$= \frac{8}{32} + \frac{10}{32} = \frac{18}{32}$$

③ $P(Y \leq 3) = P(Y=1) + P(Y=2) + P(Y=3)$

$$= \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

④ $P(X < 3, Y \leq 4) = P(X=0, Y \leq 4) + P(X=1, Y \leq 4) + P(X=2, Y \leq 4)$

$$= \left(\frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right)$$

$$= \frac{3}{32} + \frac{6}{16} + \frac{6}{64} = \frac{36}{64}$$

Independent RVs:

If (X, Y) is a two-dimensional discrete RV

$$\text{i.e. } P(X=x_i, Y=y_j) = P(X=x_i)P(Y=y_j)$$

$$\text{i.e. } \frac{P_{ij}}{P_{i*}} = \frac{P_{*j}}{P_{**}}$$

$$\text{i.e. } P_{ij} = P_{i*} \times P_{*j} \quad \forall i, j$$

then X & Y are said to be independent RVs.

For Continuous RVs

$$f(x, y) = f_x(x) \times f_y(y) \quad [\text{i.e. joint probability = product of marginal probs}]$$

then X and Y are said to be independent RVs.

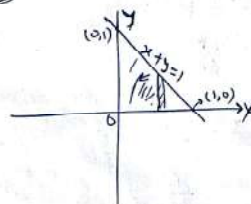
Ex The random variable X & Y have the joint density function

$$f(x, y) = \begin{cases} 6(1-x-y) & \text{for } x > 0, y > 0, x+y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

① Find the marginal distribution of X and Y .

② Are X and Y independent?

Soln The domain of defn of RVs



The marginal dist. of X

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_0^{1-x} 6(1-x-y) dy = 6 \int_0^{1-x} (1-x-y) dy \\ &= 6 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x} \\ &= 6 \left[(1-x)(1-x) - \frac{(1-x)^2}{2} \right] = 3(1-x)^2 \end{aligned}$$

marginal distribution of Y

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x,y) dx = \int_0^{1-y} 6(1-x-y) dx \\ &= 3(1-y)^2 \end{aligned}$$

To check if RVs are independent

$$\begin{aligned} f_X(x) \times f_Y(y) &= 3(1-x)^2 \times 3(1-y)^2 \\ &= 9(1-x)^2(1-y)^2 \\ &\neq f(x,y) \end{aligned}$$

$\Rightarrow$ RVs are not independent.

Ex

Given the joint probability function

$$f(x,y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal density?

conditional probability?

Are X & Y independent?

Find $P(X < \frac{1}{2})$ and $P(X < \frac{1}{2}, Y < \frac{1}{2})$

Sol

check if given function is valid probability function or not

$$\iint_{\mathbb{R}^2} f(x,y) dx dy = 1$$

$$\begin{aligned} \int_0^1 \int_0^1 4xy dx dy &= 4 \int_0^1 \int_0^1 xy dx dy = 4 \int_0^1 x \left[\frac{y^2}{2} \right]_0^1 dx \\ &= 4 \int_0^1 x \left(\frac{1}{2} - 0 \right) dx = 4 \left[\frac{x^2}{2} \right]_0^1 \\ &= 1 \end{aligned}$$

$\rightarrow$ Valid pdf

For marginal probabilities

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 4xy dy = 4x \left[\frac{y^2}{2} \right]_0^1 = 2x$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^1 4xy dx = 4y \left[\frac{x^2}{2} \right]_0^1 = 2y$$

conditional probability

$$f(x/y) = \frac{f(x,y)}{f_Y(y)} = \frac{4xy}{2y} = 2x$$

$$f(y/x) = \frac{f(x,y)}{f_X(x)} = \frac{4xy}{2x} = 2y$$

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For independence of X & Y

$$f(x,y) = f(x) \cdot f(y)$$

$$\begin{aligned} \text{i.e. } 4xy &= 2x \cdot 2y \\ &= 4xy \end{aligned}$$

$\Rightarrow$ X & Y are independent

$$\begin{aligned} P(X < \frac{1}{2}) &= \int_0^{1/2} \left\{ \int_0^1 4xy dy \right\} dx \\ &= \int_0^{1/2} f_X(x) dx = \int_0^{1/2} 2x dx = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} P(X < \frac{1}{2}, Y < \frac{1}{2}) &= \int_0^{1/2} \int_0^{1/2} 4xy dx dy = \int_0^{1/2} 4y \left[\frac{x^2}{2} \right]_0^{1/2} dy \\ &= \int_0^{1/2} 4y \left[\frac{(\frac{1}{2})^2}{2} - 0 \right] dy \\ &= \frac{1}{16} \end{aligned}$$

Mathematical Expectation of two dimensional RV

If X and Y are two RV. having joint p.m.f (pdf) $f_{X,Y}(x,y)$ then expectations of $\phi(x,y)$ is given by

(i) For discrete

$$E[\phi(x,y)] = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \phi(x_i, y_j) f_{ij}$$

(ii) For continuous

$$E[\phi(x,y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(x,y) f(x,y) dx dy$$

Exonpl

Let X takes two values x_1 & x_2 and Y only one value y ,

$$\phi(x,y) = 2x^2 + xy$$

$$E[\phi(x,y)] = E[2x^2 + xy]$$

$$= \sum_{x=1}^2 \sum_{y=1}^1 (2x_i^2 + x_i y_i) f_{X,Y}(x_i, y_i)$$

$$= \sum_{i=1}^2 [(2x_i^2 + x_i y_i) f_{X,Y}(x_i, y_i)]$$

where $f_{X,Y}(x_i, y_i) = P(X=x_i, Y=y_i)$ Probability mass function

$$\rightarrow = (2x_1^2 + x_1 y_1) f(x_1, y_1) + (2x_2^2 + 2x_2 y_1) f(x_2, y_1)$$

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Covariance

Let X and Y be two RVs

and let $E(X) = \mu_1$ and $E(Y) = \mu_2$

The quantity $E\{[X-E(X)][Y-E(Y)]\}$

is the measure of variation of two RVs from their respective means and is called covariance of X & Y .

$$\text{COV}(X, Y) = E\{[X-E(X)][Y-E(Y)]\}$$

Also called the product moment of X & Y

$$E\{[X-E(X)][Y-E(Y)]\} = E\{(X-\mu_1)(Y-\mu_2)\}$$

$$= E\{XY - \mu_2 X - \mu_1 Y + \mu_1 \mu_2\}$$

$$= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1 \mu_2$$

$$= E(XY) - \mu_2 \mu_1 - \mu_1 \mu_2 + \mu_1 \mu_2$$

$$= E(XY) - \mu_1 \mu_2$$

$$= E(XY) - E(X)E(Y)$$

$$\text{COV}(X, Y) = E(XY) - E(X)E(Y)$$

* If X and Y are independent then $E(XY) = E(X)E(Y)$

$$\Rightarrow \text{COV}(X, Y) = 0$$

$$\text{COV}(X, Y) = \frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i$$

Properties of Covariance

Let X, Y & Z be any RVs and a, b, c, d any constants. Then

$$(i) \text{COV}(X, X) = \text{Var}(X)$$

$$(ii) \text{COV}(X, Y) = \text{COV}(Y, X)$$

$$(iii) \text{COV}(aX, bY) = ab \text{COV}(X, Y)$$

$$(iv) \text{COV}(X+a, Y+b) = \text{COV}(X, Y)$$

$$(v) \text{COV}\left(\frac{X-\bar{X}}{\sigma_X}, \frac{Y-\bar{Y}}{\sigma_Y}\right) = \frac{1}{\sigma_X \sigma_Y} \text{COV}(X, Y)$$

$$(vi) \text{COV}(aX+b, cY+d) = ac \text{COV}(X, Y)$$

$$(vii) \text{COV}(X, Y+Z) = \text{COV}(X, Y) + \text{COV}(X, Z)$$

$$(viii) \text{COV}(X+Y, Z) = \text{COV}(X, Z) + \text{COV}(Y, Z)$$

Ex

Three coins are tossed. Let X denote the no. of heads on the first two and Y denote the no. of heads on the last two coins.

Find (1) the joint distribution of X and Y

(2) Marginal distribution of X & Y .

(3) $E(X)$ & $E(Y)$

(4) $\text{COV}(X, Y)$

Soln

The possible outcomes in tossing three coins

$$S = \{HHH, HHT, HTH, HTT, THT, TTH, TTT\}$$

	HHH	HHT	HTT	TTT	HTH	THH	THT	TTH
X	2	2	1	0	1	1	1	0
Y	2	1	0	0	1	2	1	1

(i) For joint probability distribution and marginal distribution

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X \ Y	0	1	2	$f_X(x)$
0	$\frac{1}{8}$	$\frac{1}{8}$	0	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
2	0	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
$f_Y(y)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1

marginal dist. of Y

(iii) $E(X)$ & $E(Y)$

$$E(X) = \sum_{i=1}^3 x_i f_X(x_i) = 0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} = 1$$

$$E(Y) = \sum_{i=1}^3 y_i f_Y(y_i) = 0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} = 1$$

(iv) $\text{COV}(X, Y) = E(XY) - E(X)E(Y)$

$$E(XY) = \sum_y \sum_x xy f_{XY}(x, y)$$

$$= \sum_y \{0 \cdot y f_{XY}(0, y) + 1 \cdot y f_{XY}(1, y) + 2 \cdot y f_{XY}(2, y)\}$$

$$= \sum_y \{y f_{XY}(1, y) + 2y f_{XY}(2, y)\}$$

$$= 0 f_{XY}(1, 0) + 1 \cdot 1 f_{XY}(1, 1) + 2 \cdot 1 f_{XY}(2, 1)$$

$$+ 2 \cdot 2 f_{XY}(2, 2) + 2 \cdot 0 f_{XY}(2, 0) + 2 \cdot 1 f_{XY}(2, 1)$$

$$+ 2 \cdot 2 f_{XY}(2, 2)$$

$$= \frac{2}{8} + 2 \times \frac{1}{8} + 2 \times \frac{1}{8} + 4 \times \frac{1}{8} = \frac{5}{4}$$

$$\text{COV}(X, Y) = \frac{5}{4} - 1 = \frac{1}{4}$$

Conditional Expectations

Discrete RV.

Conditional Expectation of a function $\phi(X, Y)$ for given $Y = y_j$

$$E\{\phi(X, Y) | Y = y_j\} = \sum_{i=1}^{\infty} \phi(x_i, y_j) P(X = x_i | Y = y_j)$$

$$= \sum_{i=1}^{\infty} \left[\phi(x_i, y_j) \frac{P(X = x_i, Y = y_j)}{P(Y = y_j)} \right]$$

$$= \frac{\sum_{i=1}^{\infty} \phi(x_i, y_j) P(X = x_i, Y = y_j)}{P(Y = y_j)}$$

Continuous RV

$$E\{g(X, Y) | Y = y\} = \int_{-\infty}^{\infty} g(x, y) f_{X|Y}(x|y) dx$$

$$= \int_{-\infty}^{\infty} g(x, y) \frac{f(x, y)}{f_Y(y)} dx$$

$$= \frac{\int_{-\infty}^{\infty} g(x, y) f(x, y) dx}{f_Y(y)}$$

gn the last example

$$E(Y | X=1) = ?$$

$$E(Y | X=1) = \sum_{y=0,1,2} y P(Y=y | X=1)$$

$$= \sum_{y=0,1,2} y \frac{P(Y=y, X=1)}{P(X=1)} = \frac{1}{\frac{1}{4}} \sum_{y=0,1,2} 2y P(X=1, Y=y)$$

$$= 2 \sum_{y=0,1,2} y P(X=1, Y=y)$$

$$= 2 [0 \times P(X=1, Y=0) + 1 \times P(X=1, Y=1) + 2 \times P(X=1, Y=2)]$$

$$= 2 \left[\frac{2}{8} + 2 \times \frac{1}{8} \right] = 1$$

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Moment Generating Function (m.g.f.)

The m.g.f. of a RV X about origin having the probability function $f(x)$ is given by

$$M(t) = E(e^{tx}) = \sum_{-\infty}^{\infty} e^{tx} p_x \quad \text{for discrete RV}$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx \quad \text{for Continuous}$$

The integration or summation being extended to the entire range of x , t being the real parameter and it is being assumed that R.H.S. is absolutely convergent.

$$M(t) = E(e^{tx}) = E\left(1 + tx + \frac{t^2 x^2}{2!} + \dots + \frac{t^r x^r}{r!} + \dots\right)$$

$$= 1 + t \mu_1' + \frac{t^2}{2!} \mu_2' + \dots + \frac{t^r}{r!} \mu_r' + \dots \quad (1)$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r'$$

$$\text{where } \mu_1' = E(X)$$

i.e. 1st moment about origin

$$\mu_2' = E(X^2)$$

i.e. 2nd moment about origin

Similarly μ_r' is the r th moment about the origin

$\Rightarrow \frac{t^r}{r!}$ denoting the r th moment about the origin that is why it is known as moment generating function.

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Differentiating eqn (1) w.r.t t , r times,

$$\frac{d^r}{dt^r} \{M(t)\} = \frac{1^r}{1^r} \mu_1' + \frac{1^r+1}{1! 1^r+1} t \mu_{r+1}' + \frac{1^r+2}{2! 1^r+2} t^2 \mu_{r+2}' -$$

put $t=0$

$$\left[\frac{d^r}{dt^r} \{M(t)\} \right]_{t=0} = \mu_r'$$

$$\therefore \mu_r' = \left[\frac{d^r}{dt^r} \{M(t)\} \right]_{t=0} = E(X^r)$$

* In general, the m.g.f. of RV X about the point 'a' is given by

$$M(t) = E[e^{t(x-a)}]$$

* If m.g.f. of X is $M_X(t)$ and if $Y = aX + b$, then $M_Y(t) = e^{bt} M_X(at)$

* If X & Y are independent RVs and $Z = X + Y$ then $M_Z(t) = M_X(t) M_Y(t)$

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Ex

Given the following discrete probability distribution

$$P[X=x] = \frac{1}{2^x}, \quad x=1, 2, \dots$$

Find the moment generating function

Soln

$$M(t) = E(e^{tx}) = \sum_{x=1}^{\infty} e^{tx} P[X=x]$$

$$= \sum_{x=1}^{\infty} e^{tx} \frac{1}{2^x} = \sum_{x=1}^{\infty} \left(\frac{e^t}{2}\right)^x$$

$$= \left(\frac{e^t}{2}\right) + \left(\frac{e^t}{2}\right)^2 + \dots$$

$$= \frac{e^t}{2} \left[1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \dots \right]$$

$$= \frac{e^t}{2} \left[1 - \frac{e^t}{2} \right]^{-1}$$

$$= \frac{e^t}{2} \cdot \left[\frac{2}{2 - e^t} \right] = \frac{e^t}{2 - e^t}$$

$$M(t) = \frac{e^t}{2 - e^t}$$

Ex

The probability density function of a RV X is as follows

$$f(x) = \begin{cases} e^{-x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the MGF

$$\text{Soln} \quad M(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} e^{-x} dx, \quad t < 1$$

$$= \int_0^{\infty} e^{-x(1-t)} dx = \frac{1}{1-t} \left[-e^{-x(1-t)} \right]_0^{\infty} = \frac{1}{1-t} \times 1 = \frac{1}{1-t}$$

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Ex The probability mass function of a RV X .

$$P[X=x] = 2\left(\frac{1}{3}\right)^x, \quad x=1, 2, 3, \dots$$

Find the MGF, μ_1' , μ_2' and $V(X)$

Soln

$$\begin{aligned} M(t) &= \sum_{x=1}^{\infty} 2\left(\frac{1}{3}\right)^x e^{tx} = 2 \sum_{x=1}^{\infty} \left(\frac{e^t}{3}\right)^x \\ &= 2 \left[\frac{e^t}{3} + \left(\frac{e^t}{3}\right)^2 + \dots \right] \\ &= \frac{2e^t}{3} \left[1 + \frac{e^t}{3} + \left(\frac{e^t}{3}\right)^2 + \dots \right] \\ &= \frac{2e^t}{3} \left(1 - \frac{e^t}{3}\right)^{-1} = \frac{2e^t}{3} \times \frac{3}{3-e^t} = \frac{2e^t}{3-e^t} \end{aligned}$$

For finding moment

$$\begin{aligned} \frac{d}{dt} M(t) &= 2 \left[\frac{(3-e^t)e^t + e^{2t}}{(3-e^t)^2} \right] = 2 \left[\frac{3e^t - e^{2t} + e^{2t}}{(3-e^t)^2} \right] \\ &= \frac{6e^t}{(3-e^t)^2} \end{aligned}$$

$$\mu_1' = \left[\frac{d}{dt} M(t) \right]_{t=0} = \frac{6}{(3-1)^2} = \frac{6}{4} = \frac{3}{2}$$

$$\begin{aligned} \mu_2' &= \left[\frac{d^2}{dt^2} M(t) \right]_{t=0} = 6 \left[\frac{(3-1)^2 + 2(3-1)}{(3-1)^4} \right] = 6 \left[\frac{4+4}{2^4} \right] \\ &= \frac{48}{16} = 3 \end{aligned}$$

$$V(X) = \mu_2' - (\mu_1')^2 = 3 - \frac{9}{4} = \frac{3}{4}$$

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Ex For a given MGF, $M(t) = \frac{1}{1-t^2}$, find the mean and variance of X .

Soln

$$\frac{d}{dt} M(t) = \frac{2t}{1-t^2}$$

$$\mu_1' = \left[\frac{d}{dt} M(t) \right]_{t=0} = 0$$

$$\frac{d^2}{dt^2} M(t) = 2 \left[\frac{1+t^2}{(1-t^2)^2} \right]$$

$$\left[\frac{d^2}{dt^2} M(t) \right]_{t=0} = 2 = \mu_2'$$

$$V(X) = \mu_2' - (\mu_1')^2 = 2$$

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Continuous Random Variable

If X is a continuous RV which can take all values (i.e. infinite no. of values) in an interval, then X is called a continuous RV.

Ex Length of time during which a vacuum tube installed in a circuit function is a continuous RV.

Probability Density function

If X is a continuous RV s.t.

$$P\left\{x - \frac{1}{2}dx \leq X \leq x + \frac{1}{2}dx\right\} = f(x)dx$$

then $f(x)$ is called the probability density function (pdf) of X , provided $f(x)$ satisfies following conditions

(i) $f(x) \geq 0$, $\forall x \in R_x$, and

$$(ii) \int_{R_x} f(x)dx = 1$$

moreover, $P(a \leq X \leq b)$ or $P(a < X < b)$ is defined as

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

The curve $y = f(x)$ is called the probability curve of RV X .

* When X is a continuous RV

$$P(X=a) = P(a \leq X \leq a) = \int_a^a f(x)dx = 0$$

$\Rightarrow$ It is almost impossible that a continuous RV assumes a specific value.

Hence

$$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$$

Cumulative Distribution Function (cdf)

If X is an RV, discrete or continuous then $P(X \leq x)$ is called the cumulative distribution function of X or distribution of X and denoted as $F(x)$

If X is discrete

$$F(x) = \sum_{x_j \leq x} p_j$$

If X is continuous

$$F(x) = P(-\infty < X \leq x) = \int_{-\infty}^x f(x) dx$$

Properties of cdf $F(x)$

- ① $F(x)$ is a non-decreasing function of x , i.e.
if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$
- ② $F(-\infty) = 0$ and $F(\infty) = 1$
- ③ If X is a discrete RV taking values $x_1, x_2, \dots$
where $x_1 < x_2 < x_3 < \dots < x_{i-1} < x_i < \dots$
then $P(X = x_i) = F(x_i) - F(x_{i-1})$
- ④ If X is a continuous RV, then $\frac{d}{dx} F(x) = f(x)$,
at all points where $F(x)$ is differentiable.

Ex

$$f(x) = 6x(1-x), \quad 0 \leq x \leq 1$$

check if $f(x)$ is probability density function

Soln

For Pdf

$$\begin{aligned}\int_0^1 f(x) dx &= \int_0^1 6x(1-x) dx \\&= 6 \left[\int_0^1 (x-x^2) dx \right] \\&= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 6 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{6}{6} = 1 \\&\Rightarrow f(x) \text{ is Pdf.}\end{aligned}$$

Ex

Find the value of k for the pdf

$$f(x) = \begin{cases} kx^2 & \text{where } 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

also compute $P(1 \leq x \leq 2)$ and the distributive function

Soln

$\therefore f(x)$ is Pdf

$$\Rightarrow \int_0^3 f(x) dx = 1$$

$$\Rightarrow \int_0^3 kx^2 dx = 1$$

$$\Rightarrow k \left(\frac{x^3}{3} \right)_0^3 = 1$$

$$\Rightarrow k \times 9 = 1$$

$$\Rightarrow k = \frac{1}{9}$$

$$P(1 \leq x \leq 2) = \int_1^2 f(x) dx = \int_1^2 kx^2 dx = \int_1^2 \frac{x^2}{9} dx = \frac{7}{27}$$

For distributive function

$$F(x) = \int_0^x f(x) dx = \int_0^x kx^2 dx = \int_0^x \frac{x^2}{9} dx$$
$$= \left[\frac{x^3}{27} \right]_0^x = \frac{x^3}{27}$$

$$F(x) = \begin{cases} 0 & , x < 0 \\ \frac{x^3}{27} & 0 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

* Note - If X_1 & X_2 are RV's & c is a constant then cX_1 , $X_1 + X_2$, $X_1 X_2$ & $C_1 X_1 + C_2 X_2$ are also random variables, where C_1 & C_2 are constants.

* $X_1 - X_2$ is also RV

* If X is a RV

(i) $\frac{1}{X}$, where $\frac{1}{X} = \infty$ if $X = 0$

(ii) $X_+ = \max\{0, X\}$, (iii) $X_- = -\min\{0, X\}$

(iv) $|X|$ also RV.

* If X is RV & f is a continuous (or) increasing function then $f(X)$ is also a RV.

Mathematical Expectation of One Dimensional Random Variable

Defn If X is a discrete RV, then the expected value or mean value of $g(x)$ is defined as

$$E\{g(x)\} = \sum_i g(x_i) p_i$$

where $p_i = P(X=x_i)$ is the probability mass function of X .

If X is a continuous RV with pdf $f(x)$ then

$$E\{g(x)\} = \int_{R_X} g(x) f(x) dx$$

Mean, Variance, and Standard deviation of a Random Variable

Expected values mean μ_X and Variance σ_X^2

$$\mu_X = E(X)$$

$$= \sum_i x_i p_i, \text{ if } X \text{ is discrete}$$

$$= \int_{R_X} x f(x) dx, \text{ if } X \text{ is continuous}$$

$$\text{Var}(X) = \sigma_X^2 = E[(X - \mu_X)^2]$$

$$= \sum_i (x_i - \mu_X)^2 p_i, \text{ if } X \text{ is discrete}$$

$$= \int_{R_X} (x - \mu_X)^2 f(x) dx, \text{ if } X \text{ is continuous}$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

Properties of Expected values

- (i) $E(aX) = a E(X)$
- (ii) $\text{Var}(aX) = a^2 \text{Var}(X)$
- (iii) $E(X+Y) = E(X) + E(Y)$
- (iv) If X and Y are independent RVs, then

$$E(XY) = E(X) \cdot E(Y)$$

- (v) If $X \geq 0$, then $E(X) \geq 0$

Ex A random Variable X has the following probability function

$x:$	0	1	2	3	4	5	6	7
$P(X):$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

then find (i) $k = ?$, (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$,
 $P(0 < X < 5)$

(iii) $P(X \leq a) > \frac{1}{2}$ find the minimum value of a .

(iv) Determine the distribution function of X .

(v) Find mean and variance.

Soln (i) $\therefore \sum p_i = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$\Rightarrow 10\left(k - \frac{1}{10}\right)(k+1) = 0$$

$$\Rightarrow k = \frac{1}{10} \text{ or } -1 \text{ (not possible)}$$

$$\therefore k = \frac{1}{10}$$

(iv) Distribution table with distribution function

x	0	1	2	3	4	5	6	7
$P(x)$	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$
$F(x) = P(X \leq x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{83}{100}$	1

$$\begin{aligned} \text{(ii)} \quad P(X < 6) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + \\ &\quad P(X=4) + P(X=5) \\ &= 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \frac{81}{100} \end{aligned}$$

$$P(X \geq 6) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$\begin{aligned} P(0 < X < 5) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} = \frac{8}{10} = 0.8 \end{aligned}$$

(iii) $P(X \leq a) > \frac{1}{2}$ where a is minimum

$$P(X \leq 3) = \frac{5}{10} = 0.5$$

$$P(X \leq 4) = \frac{8}{10} = 0.8 > \frac{1}{2}$$

$$\therefore a = 4$$

(v) mean = $E(X) = \sum x_i p_i$

$$= \frac{1}{10} + \frac{4}{10} + \frac{6}{10} + \frac{12}{10} + \frac{5}{100} + \frac{12}{100} + \frac{119}{100}$$

$$= \frac{366}{100} = 3.66$$

$$E(X^2) = \sum_i x_i^2 p_i$$

$$= \frac{1}{10} + \frac{8}{10} + \frac{18}{10} + \frac{48}{10} + \frac{25}{100} + \frac{72}{100} + \frac{833}{100}$$

$$= 16.8$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 16.8 - (3.66)^2$$

$$= 16.8 - 13.4 = \underline{3.4}$$

Ex If the density function of a continuous R.V. X is given by

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

(1) Find value of a

(2) Find the cdf of X

Soln

$$\int_0^3 f(x) dx = 1$$

$$\int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (3a - ax) dx = 1$$

$$a \left[\frac{x^2}{2} \right]_0^1 + a[x]_1^2 + \left[3ax - a \frac{x^2}{2} \right]_2^3 = 1$$

$$\frac{a}{2} + a + \left[9a - \frac{9}{2}a - 6a + 2a \right] = 1$$

$$a \left(\frac{1}{2} + 1 + \frac{1}{2} \right) = 1 \Rightarrow a = \frac{1}{2}$$

(ii) cdf of X

$$F(x) = P(X \leq x) = 0 \quad \text{when } x < 0$$

$$F(x) = \int_0^x \frac{x}{2} dx = \frac{x^2}{4}, \quad \text{when } 0 \leq x < 1$$

For $1 \leq x < 2$

$$\begin{aligned} F(x) &= \int_0^1 \frac{x}{2} dx + \int_1^x \frac{1}{2} dx = \frac{1}{2} \left(\frac{x^2}{2} \right)_0^1 + \frac{1}{2} (x)_1^x \\ &= \frac{1}{2} \left(\frac{1}{2} - 0 \right) + \frac{1}{2} (x - 1) = \frac{x}{2} - \frac{1}{4} \\ &= \frac{x}{2} - \frac{1}{4} \end{aligned}$$

For $2 \leq x < 3$

$$\begin{aligned} F(x) &= \int_0^1 \frac{x}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^x \left(\frac{3}{2} - \frac{x}{2} \right) dx \\ &= \frac{3}{2} x - \frac{x^2}{4} - \frac{5}{4} \end{aligned}$$

For $x > 3$, $F(x) = 1$

Ex 9) X is a CRV with the following pdf

$$f(x) = \begin{cases} \alpha(2x - x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find (i) α

(ii) $P(X > 1)$

Ex The PDF of random variable X is

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

① Find $P(X \geq 1.5)$

② Find Cdf

Soln ① $P(X \geq 1.5) = \int_{1.5}^{\infty} f(x) dx = \int_{1.5}^2 (2-x) dx = 0.125$

② ~~Find~~ For Cdf

g1 $x \leq 0$

$$F(x) = \int_{-\infty}^x f(x) dx = 0$$

g1 $0 < x < 1$

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx = 0 + \int_0^x x dx = \frac{x^2}{2}$$

g1 $1 \leq x < 2$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx \\ &= 0 + \int_0^1 x dx + \int_1^x (2-x) dx = \left(\frac{x^2}{2}\right)_0^1 + \left(2x - \frac{x^2}{2}\right)_1^x \\ &= -1 + 2x - \frac{x^2}{2} \end{aligned}$$

For $x \geq 2$

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^x f(x) dx = 1$$

$$F_n(x) = \begin{cases} 0 & -\infty < x < 0 \\ x^2/2 & x \leq 1 \\ -1 + 2x - \frac{x^2}{2} & x \leq 2 \\ 1 & > 2 \end{cases}$$

Q2

Find mean and variance of the probability distribution given by following table

x	1	2	3	4	5
$P(x)$	0.2	0.35	0.25	0.15	0.05

$$E(X) = \sum xp(x) = 0.2 + 0.7 + 0.75 + 0.6 + 0.25$$

$$\text{mean} = 2.5$$

$$E(X^2) = \sum x^2 p = 0.2 + 4 \times 0.35 + 9 \times 0.25 + 16 \times 0.15 + 25 \times 0.05$$

$$= 7.5$$

$$\text{Var}(X) = \sigma_x^2 = E(X^2) - (E(X))^2$$

$$= 7.5 - (2.5)^2$$

$$= 7.5 - 6.25 = 1.25$$

Q3

A continuous RV X has density function.

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find expected value & variance of X .

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x (2e^{-2x}) dx$$

$$\text{mean} = \frac{1}{2}$$

$$E(X^2) = \int_0^{\infty} x^2 (2e^{-2x}) dx = \frac{1}{2}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$